W13 · THE FOURTH ARABIC NLP CONFERENCE (ArabicNLP 2026) · EMNLP 2026 · BUDAPEST

ARABIC SENTIMENT SWAP · DIALECTSENTEVAL 2026 SUBTASK 2

Retrieval-Augmented, Preference-Tuned Generation

for Arabic Sentiment Swap

Momen Ibrahim · Mariam E. Hassib · Ahmed Salem · Marwan Torki

Computer and Systems Engineering Department, Faculty of Engineering, Alexandria University, Egypt | Team ALEXIS

Rank #1 of 15 submissions · SSA 0.9581 · BLEU 56.91 · chrF 72.18

RESULTS AT A GLANCE

First of fifteen, by 17.3 points of style accuracy

0.9581

Sentiment Style Accuracy

official test set, rank 1 of 15

+17.3 pts

SSA ahead of 2nd place

0.9581 vs 0.7851

56.91 / 72.18

BLEU / chrF

against the human rewrites

+0.155

SSA from one fix

a dedicated source-polarity classifier

The setting

Rewrite a Modern Standard Arabic sentence so it carries the opposite polarity while its topic stays the same.

Our approach

Constrained editing: a pool of four generators plus retrieval, one score balancing polarity and content, and an explicit, well-predicted target polarity.

THE TASK

Flip the polarity, keep everything else

Two objectives pull against each other: the output must land firmly on the opposite side of the polarity boundary, yet stay close to the source, because content is scored against a human rewrite that is itself a small edit.

SOURCE

تجربة **جميلة** سعر كتير مناسب شكرا **لكم**

‘a nice experience, reasonable price, thank you’

REWRITE

تجربة **سيئة** سعر كتير **غير** مناسب **لا** شكرا **عليكم**

‘a bad experience, unreasonable price, no thanks’

A training pair from MA’AKS: a few words change (coloured); the structure and length do not.

MA’AKS: Modern Standard Arabic swap pairs

MA’AKS	Train	Val	Test
Pairs	6,752	964	1,931
Unique sources	5,517	–	–
Has source_polarity	**Yes**	**Yes**	**No**
Exact duplicate of a train source	–	20.4%	22.8%

Unlike train and validation, the test file carries no source_polarity column.

3,731 / 3,021

negative / positive training sources

60.7–63.7%

of sentences contain emoji

How systems are scored

SSA Sentiment Style Accuracy: the fraction of outputs a fixed CAMeLBERT sentiment classifier assigns to the target polarity.

BLEU n-gram overlap with the human rewrite.

chrF character n-gram F-score against the human rewrite.

Three data properties, three components

91.0%

of training pairs stay within two words of the source's length; character similarity has median 0.82 (IQR 0.75–0.88)

Local edit, not paraphrase

An edit-tagging pool member and a chrF content term in the reranker

22.8%

of test sentences are byte-identical to a train or validation source

Retrieval index

A gold rewrite already exists: a strong candidate at no modelling cost

323

aligned emoji flips; short antonym pairs such as جميلة → كريهة (115 occurrences)

Polarity-flip lexicons

A deterministic safety net for markers the neural models leave behind

THE HIDDEN CATCH

The test set omits the source polarity

The task description says it is given, and the training files contain it, but the test file does not: the target polarity has to be predicted first.

74.3%

the grading classifier, zero-shot

Roughly a quarter of the test set was being rewritten toward the wrong target, and no generator tuning could see it.

96.7%

our fine-tuned classifier

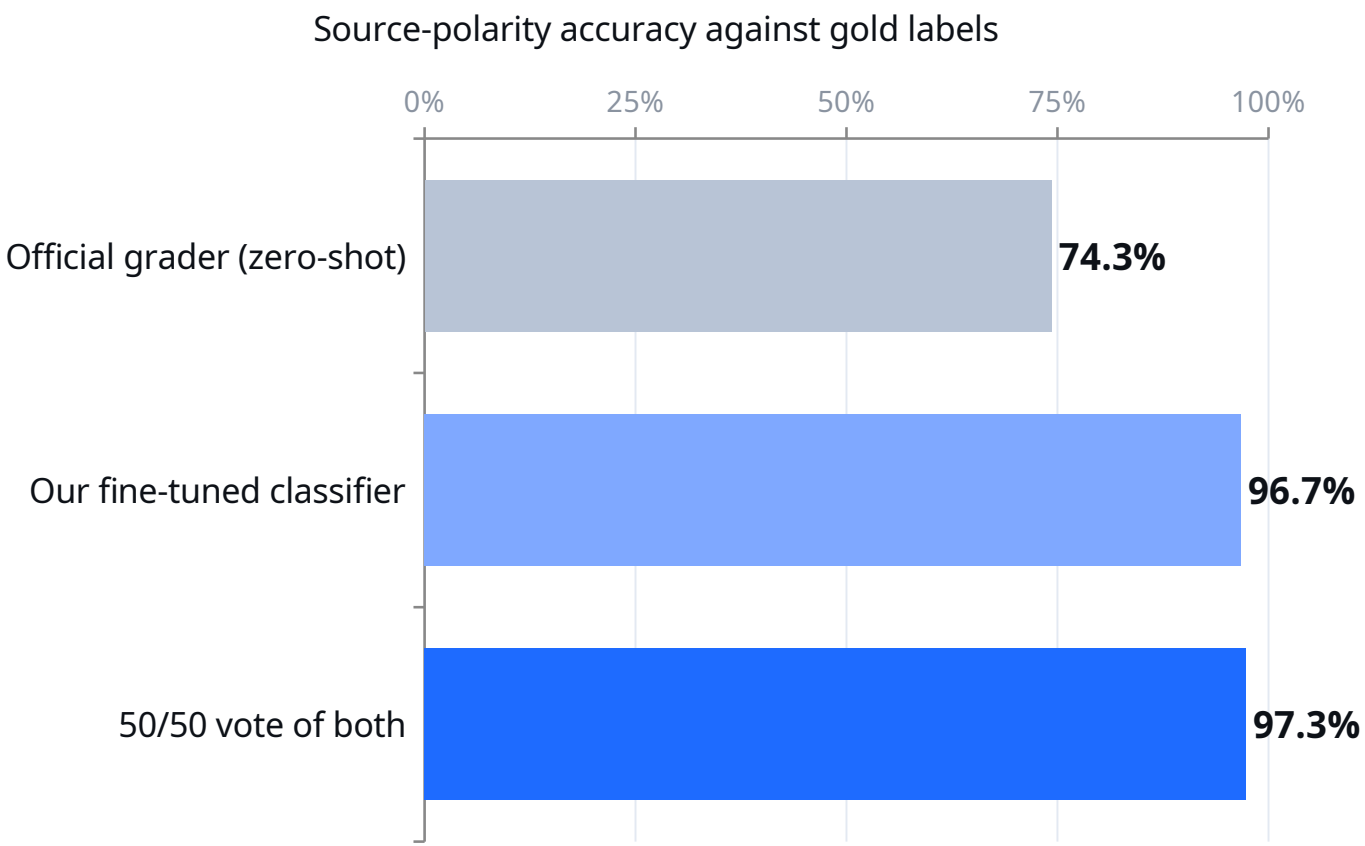
Trained on the 7,716 labeled train + validation pairs.

97.3%

a 50/50 vote of both

The two fail in different places, so the vote beats either one.

A dedicated classifier fixes the target



+0.155

SSA, our largest single gain

0.7887 → 0.9436 on the official leaderboard

Why it matters

If the source polarity is wrong, the system rewrites a negative sentence to be more negative, and no beam width or reranking weight recovers from that.

01

The System

Detect the polarity, pool candidates, rerank, repair

Three stages

1 Detect

Predict the source polarity (dedicated + official classifier vote, 97.3%). The opposite label is the target t that everything downstream aims at.

2 Candidate pool

mT5-base	10 beams; 13 if source predicted negative; fine-tuned + DPO
mT5-large	6 beams
NLLB-200 (600M)	3 beams; Arabic → Arabic “translator”
Edit tagger	KEEP / DELETE / REPLACE; reaches the reference for 90.6% of validation sources
Retrieval	gold rewrite of any training source within 0.90 cosine

3 Rerank, repair

Every candidate, retrieved ones included, is scored with one function; the best is returned. A deterministic pass then flips any emoji or antonym that still carries the source polarity.

Retrieved near-duplicates are ordinary candidates, not a bypass: they must win on the same terms as generated ones.

The same score reranks candidates and rewards DPO

$$s(y) = \alpha \cdot p_t(y) + (1 - \alpha) \cdot \mathrm{chrF}(y, x)$$

α = 0.85 when reranking · p_t: the grading classifier's raw three-way softmax probability of the target polarity · x: source, y: candidate

Reranker

Every candidate from the pool is ranked by s(y) and the top one is returned. Putting the grading classifier inside the score is deliberate: from here on, agreeing with the grader is the objective.

DPO reward

4,500 preference examples, 7 epochs: sample 4 candidates, add the gold rewrite, score all five with s(y). An early reward dropped neutral and renormalised two classes, which the model exploited; the raw three-way probability closes the loophole.

Optimising hard against an imperfect proxy degrades true performance (reward over-optimisation, Gao et al., 2023).

Details that mattered for mT5-base

Direction (validation)	chrF	SSA
Negative → positive	**78.94**	**94.37%**
Positive → negative	84.27	96.06%

Negative-to-positive is the harder direction, an asymmetry also reported for sentiment transfer more broadly.

Fully fine-tuned on all 7,716 pairs

Train + validation pairs; validation figures quoted during development come from the held-out 964-row split.

Asymmetric decoding

13 beams instead of 10 when the source is predicted negative.

Select checkpoints by beam-search chrF

Not by teacher-forced loss: the lowest-loss checkpoint wrote measurably worse text once it had to decode freely.

Arabic T5 checkpoints discarded

Two were tried: one had an unparseable tokenizer format, the other gave incoherent zero-shot output.

AI MØMEnt

02

Results

Official leaderboard, ablation and lessons

OFFICIAL TEST SET

DialectSentEval 2026 Subtask 2 leaderboard

#	Team / participant	SSA	BLEU	chrF
1	**ALEXIS (ours)**	**0.9581**	**56.91**	**72.18**
2	akarrouch	0.7851	49.82	70.12
3	ahmadalmoustafa	0.7789	41.48	64.45
4	mona_	0.7597	51.58	71.02
5	melmoussaoui	0.7530	49.03	69.53
6	mohamedsabaa	0.7447	49.07	69.33
7	hsnhb7	0.7343	46.80	67.10
8	areejkhaled	0.6898	54.63	72.61
9	mohammed_bahgat	0.6732	41.38	65.84
10	ehabashraf	0.6732	41.38	65.84
11	ahmad_alshattha	0.6354	25.07	56.33
12	hoanguitcs	0.6220	49.27	68.26
13	normalman159	0.5997	11.31	38.22
14	mustafaelkady	0.4772	0.03	10.04
15	mohamedmhaouach	0.2807	43.06	64.61

#1

of 15 submissions

first place overall

0.9581

SSA

BLEU 56.91 · chrF 72.18

+17.3

SSA points

ahead of second place (0.7851)

The margin is specific to SSA: several systems are within a few points on BLEU and chrF.

Where the gains came from

Configuration (each row an independent leaderboard submission)	SSA	Δ
mT5-base + retrieval + reranking	0.7452	–
+ Edit tagger, wider pool	0.7618	+0.017
+ DPO, NLLB-200 & mT5-large	0.7887	+0.027
+ Dedicated polarity classifier	**0.9436**	**+0.155**
+ Hard polarity gate (reverted)	**0.8933**	**−0.050**
+ Raw-softmax reranking	0.9498	+0.006
+ DPO reward fix (final system)	**0.9581**	**+0.008**

Δ against the previous retained configuration: the hard gate was reverted, so the next row is measured from 0.9436.

+0.044

Generation-side additions

tagger, wider pool, DPO, NLLB-200, mT5-large

+0.155

Polarity classifier alone

more than three times the generation-side gains

Changes that looked safe and were not

The hard gate: −0.050

Discarding candidates the classifier did not label with the target polarity appeared to cost nothing on validation, then lost 0.050 on the leaderboard: the same classifier chose and judged the candidates. A rule validated only against the signal it optimizes will always look correct.

Validation preferred α = 1

All weight on polarity and none on content. Submitted, it lowered BLEU and chrF on the leaderboard, so we moved only part of the way toward each validation optimum (α = 0.85).

Cost-aware deployment

Four generators, a tagger and a retrieval index are expensive at inference. The pooled generators add about +0.044 against +0.155 for the classifier: a latency-sensitive deployment could keep only mT5-base and the classifier.

Both regressions looked safe on validation and only showed up on the official leaderboard: score every change on the real test signal.

CONCLUSION

What to take away

Treat the swap as constrained editing

91% of rewrites are local edits: retrieval, an edit tagger and a content term keep outputs close to the source.

One score, two jobs

Polarity and content balanced in one function, used as the reranker and as the DPO reward, on the raw three-way probability.

Rank #1 of 15 submissions

SSA 0.9581, BLEU 56.91, chrF 72.18: 17.3 SSA points ahead of second place.

TAKE-HOME

The largest gain came from predicting a label the task assumed would be given.

When a shared task is built around an auxiliary classifier, measure that classifier on the sub-task it is asked to perform before trusting it.

github.com/Momen-Ibrahim-Fawzy/ALEXIS-DialectSentEval-Subtask2-SentimentSwap-

AI MØMEnt

Thank You

Questions welcome

Momen Ibrahim · es-momen.ibrahim2019@alexu.edu.eg

Team ALEXIS · Alexandria University, Egypt

PAPER
ACL Anthology

CODE
GitHub

aclanthology.org/2026.arabicnlp-sharedtasks.188
github.com/Momen-Ibrahim-Fawzy/ALEXIS-DialectSentEval-Subtask2-SentimentSwap-

Key hyperparameters of the deployed system

Hyperparameter	Value
Primary generator	mT5-base
Primary beams (standard / hard direction)	10 / 13
NLLB-200 beams (pooled)	3
mT5-large beams (pooled)	6
DPO preference examples	4,500
DPO candidates per example	4 sampled + gold
DPO epochs	7
Reranking weight α (polarity)	0.85
Reranking weight 1 − α (chrF)	0.15
Retrieval similarity threshold	≥ 0.90 (cosine)